Soham Khatavkar

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Education

Indian Institute of Technology Bombay July 2025 – Present
M.Tech in Computer Science – CGPA: 9.21/10.0 *Mumbai, Maharashtra*

National Institute of Technology Nagpur July 2021 – May 2025
B.Tech in Computer Science and Engineering – CGPA: 8.32/10.0 *Nagpur, Maharashtra*

Projects

sIDE+ | *MongoDB, ExpressJS, ReactJS, NodeJS, Azure, Docker, Gemini AI* Demo | Frontend | Backend

- Developed a full-featured **online code execution platform** supporting C, C++, Java, Python, and JavaScript.
- Integrated **Docker-based isolated execution environments** using Docker images (e.g., **GCC**, **OpenJDK**) with features like **execution time limits** and **automatic container cleanup**, to enhance security.
- Implemented features such as **auto-save**, **auto-completion**, **cloud-save**, and **code-share** within a rich code editor interface integrated with **Gemini AI** to enhance the user experience.

CrHash | *C, C++, CUDA, Make* GitHub

- Developed a **hardware-accelerated** password cracking system for MD5 and SHA-1 using **CUDA** and **C++** to evaluate billions of hashes concurrently by maximizing thread-level parallelism on the GPU.
- Implemented **brute-force** and **dictionary attack** with batched input processing to maximize GPU utilization.
- Achieved a **78.2x** speedup, sustaining a peak throughput of **5.6B hashes/sec** for MD5, and a **61.4x** speedup reaching **4.05B hashes/sec** for SHA-1 over optimized **multi-threaded** CPU baselines.

CuSync | *C++, CUDA, OpenCV, CMake* GitHub

- Developed a **high-performance video processing engine** in C++ and CUDA, implementing **custom CUDA image processing kernels** for optimized memory access patterns and thread block execution.
- Built an **asynchronous execution pipeline** via CUDA streams to overlap PCIe data transfers with GPU computation, significantly reducing processing latency and eliminating wasted hardware cycles.
- Integrated an interactive OpenCV HighGUI interface for real-time playback controls and live parameter tuning.

Setris | *C++, Raylib, Make* Demo | GitHub

- Developed a custom implementation of **Tetris** in **C++**, enabling players to fully customize gameplay rules such as speed, block shapes, and drop behavior.
- Built with a fully modular architecture leveraging core **Object-Oriented Programming (OOP)** principles, enabling clean separation of concerns, enhanced code reusability, and ease of scalability and maintenance.

Achievements

- **Winner – Microland Hackathon**; built a full-stack AI-powered health chatbot, leading to a **full-time Software Developer offer** from **Microland**.
- **Selected as Deputy Field Officer (Technical)** at the **Cabinet Secretariat**.
- **Selected as Junior Executive (IT) Engineer** at the **Airports Authority of India**.
- **Selected as Executive Engineer** at the **National Thermal Power Corporation Limited**.

Technical Skills

Languages: C, C++, CUDA, JavaScript, SQL, HTML, CSS
Frameworks & Libraries: ReactJS, NodeJS, ExpressJS, Raylib, REST APIs
Databases & Cloud: MongoDB, MySQL, Azure, Cloudinary, Render
Tools & Build Systems: Docker, CMake, Make, Git, GitHub, CUDA Toolkit, Linux

Relevant Coursework

- Data Structures
- Operating Systems
- Computer Networks
- Database Management
- OOP
- Algorithms
- Web Security
- GP-GPU